VLSI Devices
Lecture 2

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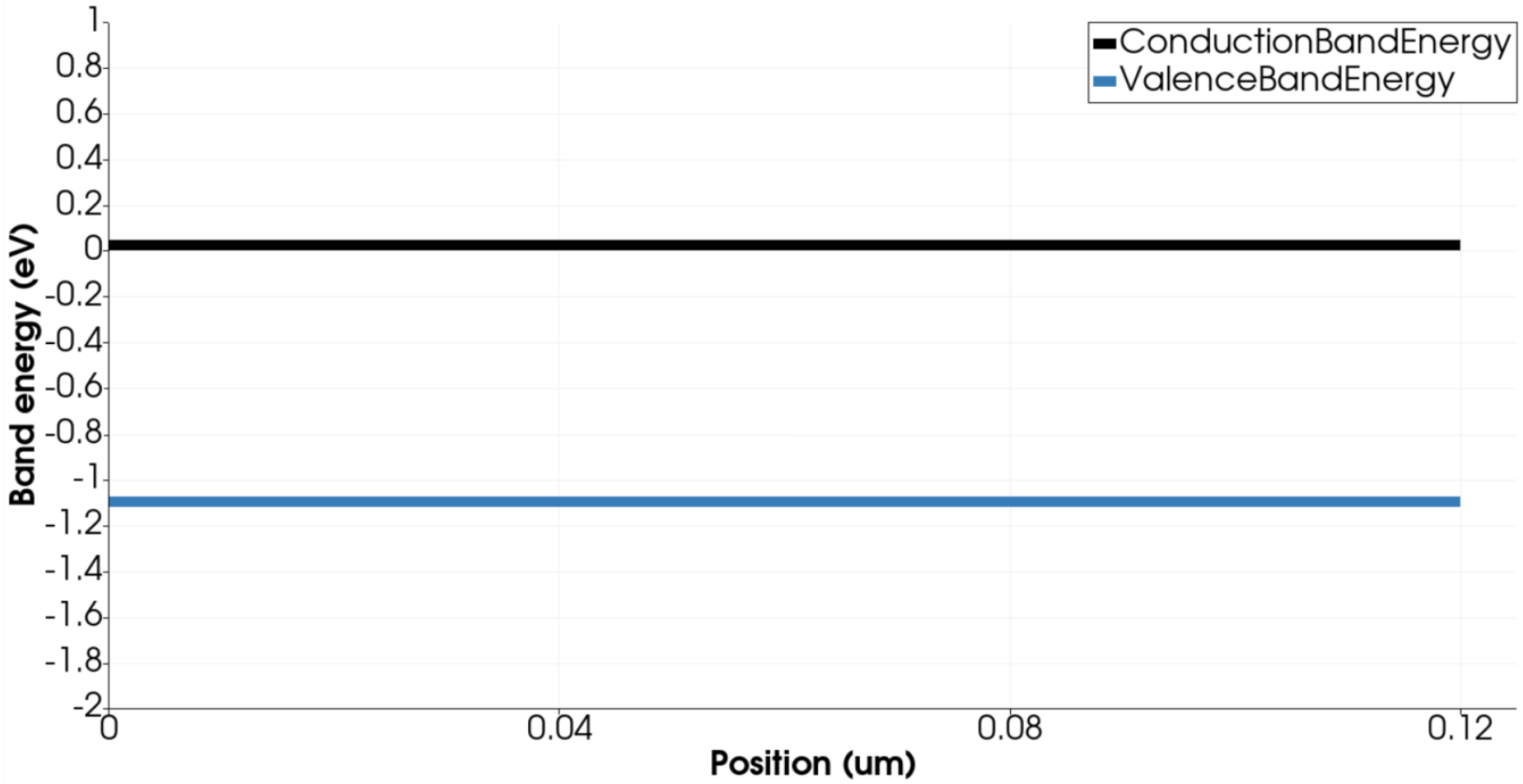
L2

A homogenous device, 10^{19} cm^{-3}

- Assume that we have a 120-nm-long homogeneous device.
  - Potential? 0.534 V



Poisson equation

- In a 1D structure,
  - The Poisson equation reads

$$\frac{d^2}{dx^2}\phi = -\frac{\rho}{\epsilon}$$

Taur, Eq. (2.56)

  - When the honomegeous device is considered, $\phi(x)$ is constant and $\rho(x)$ vanishes everywhere. ➔ The Poisson equation holds.

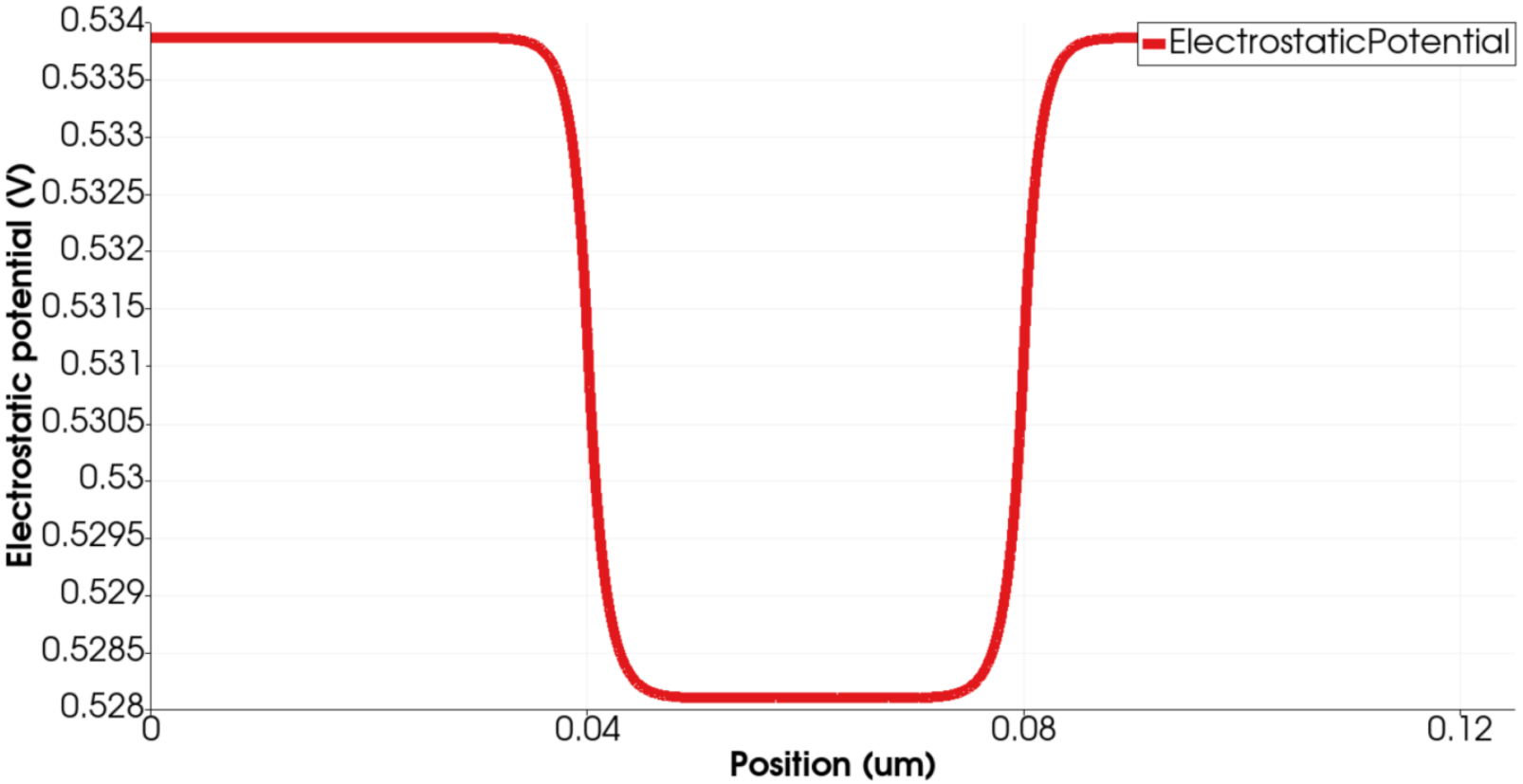
$$\rho(x) = 0 = qN_d - qn_i \exp\left(\frac{q}{k_B T}\phi_0\right)$$

  - Therefore,

$$\phi_0 = \frac{k_B T}{q}\log\left(\frac{N_d}{n_i}\right)$$

An NNN device

- Its 40-nm-long center region has a lower density, $8X10^{18}$ cm^{-3}.
  - 0.528 V @ center
  - Its transition



Poisson equation, again

- In this structure,
  - The doping density abruptly changes from N_d $(x < 0)$ to $N_d + \Delta N_d$ $(x > 0)$.
  - When $\phi(x) = \phi_0 + \Delta\phi(x)$,

$$\frac{d^2}{dx^2}\Delta\phi = \frac{q}{\epsilon}\left[n_i \exp\left(\frac{q\phi}{k_B T}\right) - N_d - \Delta N_d\right] \approx \frac{q}{\epsilon}\left[N_d \frac{q\Delta\phi}{k_B T} - \Delta N_d\right]$$

Taur, ~Eq. (2.66)

Only when $x > 0$

  - In any case, we have $\frac{d^2}{dx^2}\Delta\phi - \frac{q^2 N_d}{\epsilon k_B T}\Delta\phi = \cdots$.

Its solution

- Let us introduce the Debye length, L_D.

$$\frac{1}{L_D^2} = \frac{q^2 N_d}{\epsilon k_B T}$$

Taur, ~Eq. (2.67)

  - In this notation,

$$\frac{d^2}{dx^2}\Delta\phi - \frac{1}{L_D^2}\Delta\phi = \cdots$$

  - Its solution is a linear combination of $\exp\left(\frac{x}{L_D}\right)$ and $\exp\left(-\frac{x}{L_D}\right)$.
  - It takes a distance on the order of L_D for the silicon bands to follow an abrupt spatial change in doping concentration.
  - (At 10^{19} cm^{-3}, L_D is about 1.3 nm.)

Homework#2

- Due: 08:00 on March 11
- Submit a PDF report through the GIST LMS system.
  - Consider a homogeneous device. Calculate the electrostatic potential, when its doping concentration is 10^{19} cm^{-3}. (If not stated otherwise, assume 300 K.)

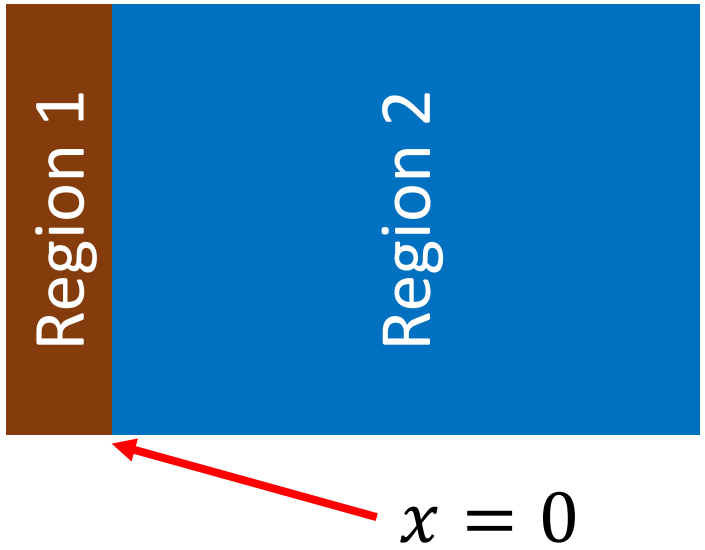
Boundary condition

- At the interface,
  - The perpendicular component of the displacement, $\mathbf{D}$, should be continuous.
  - For example, if the interface between Material 1 and Material 2 is located at $x = 0$,

$$\epsilon_1 E_x(0^-) = \epsilon_2 E_x(0^+)$$

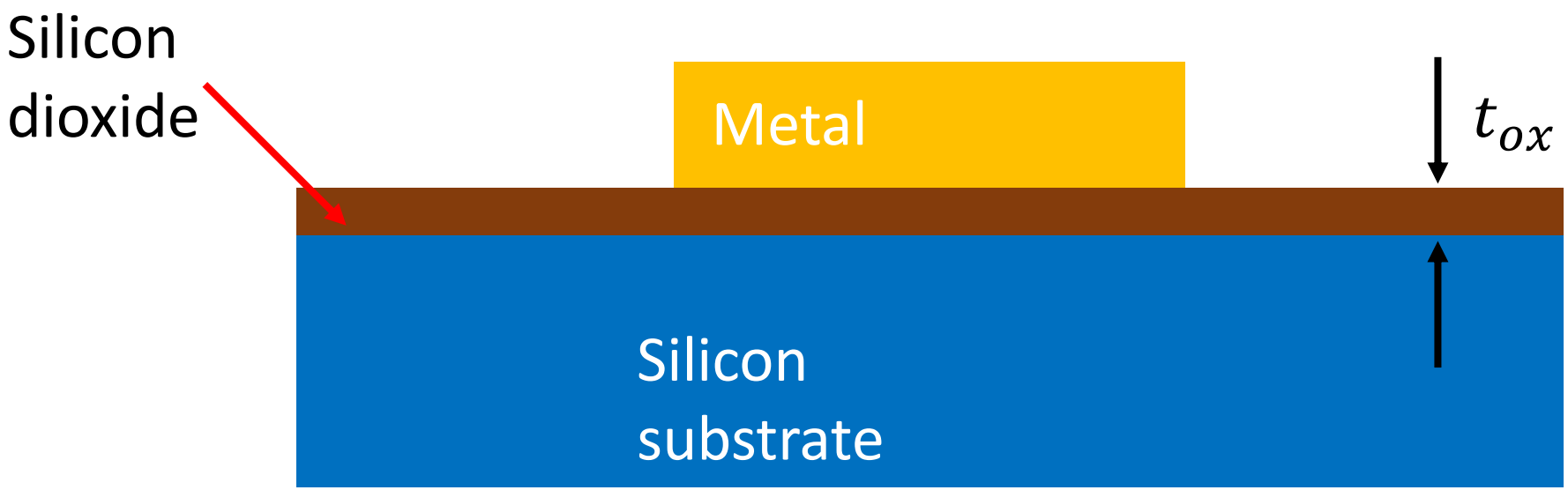
Taur, Eq. (2.61)

  - (No sheet charge at the interface)



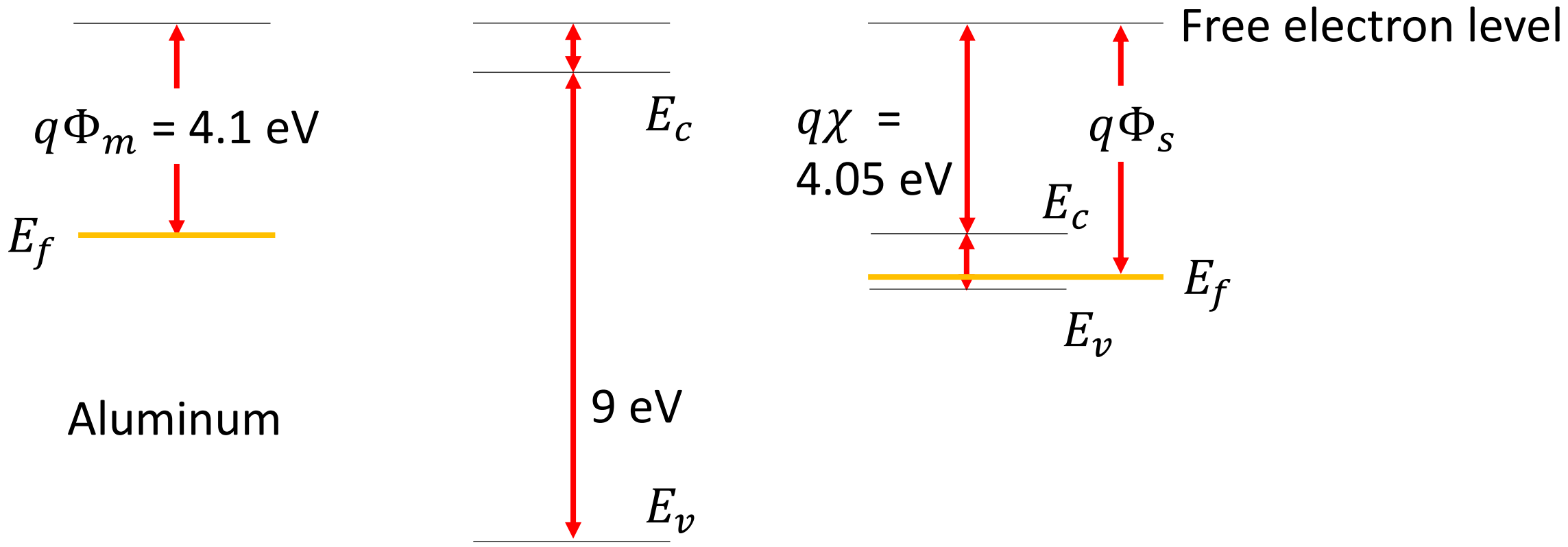
MOS capacitors

- Basis of CMOS technology



Energy band diagram

- Three components
  - Metal, silicon dioxide, and p-type silicon

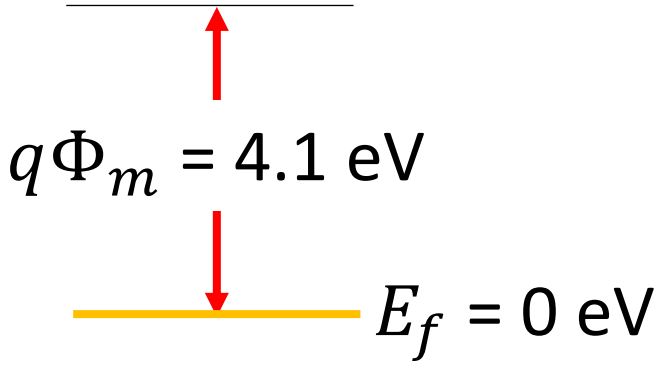


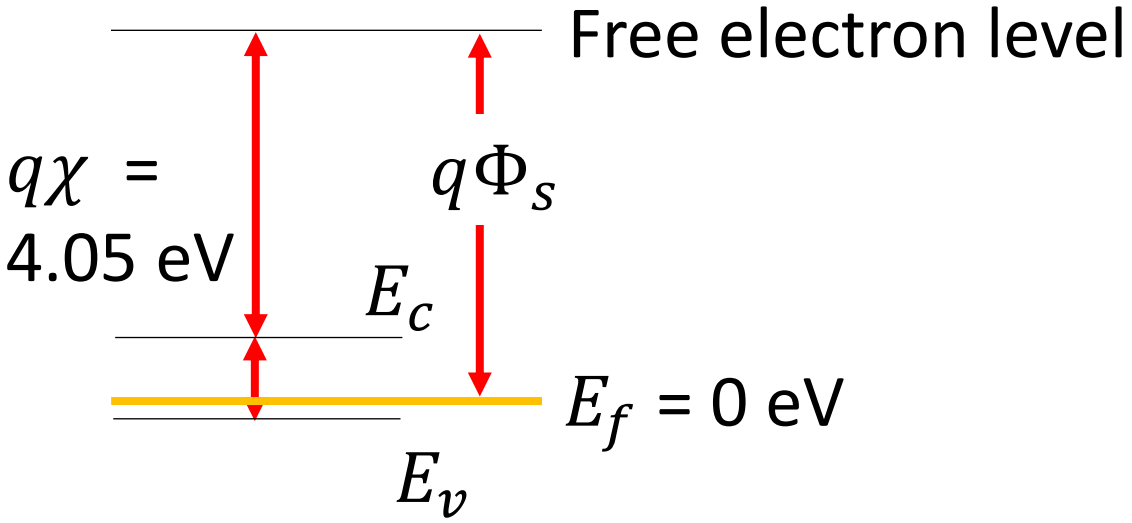
Consider $V_g = V_{sub} = 0$ V.

- Rule: Align the Fermi level.
  - The energy difference is

$$q\Phi_s - q\Phi_m$$

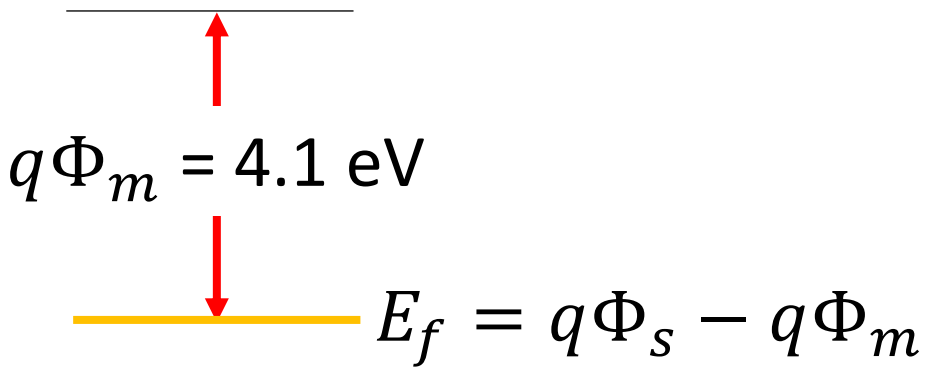
  - It means that a non-zero electric field is applied in the oxide layer.

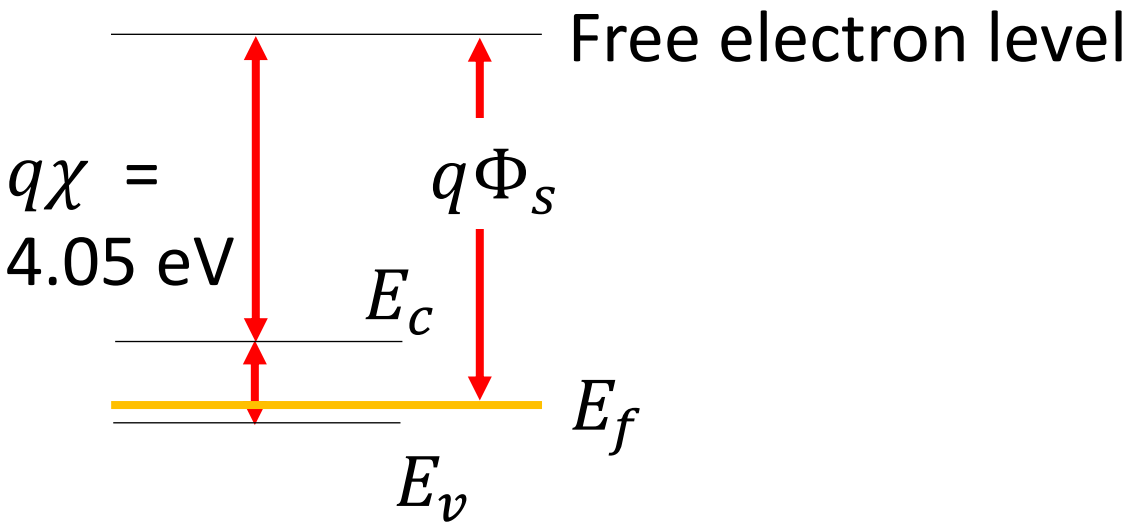




When $V_g = \Phi_m - \Phi_s < 0$,

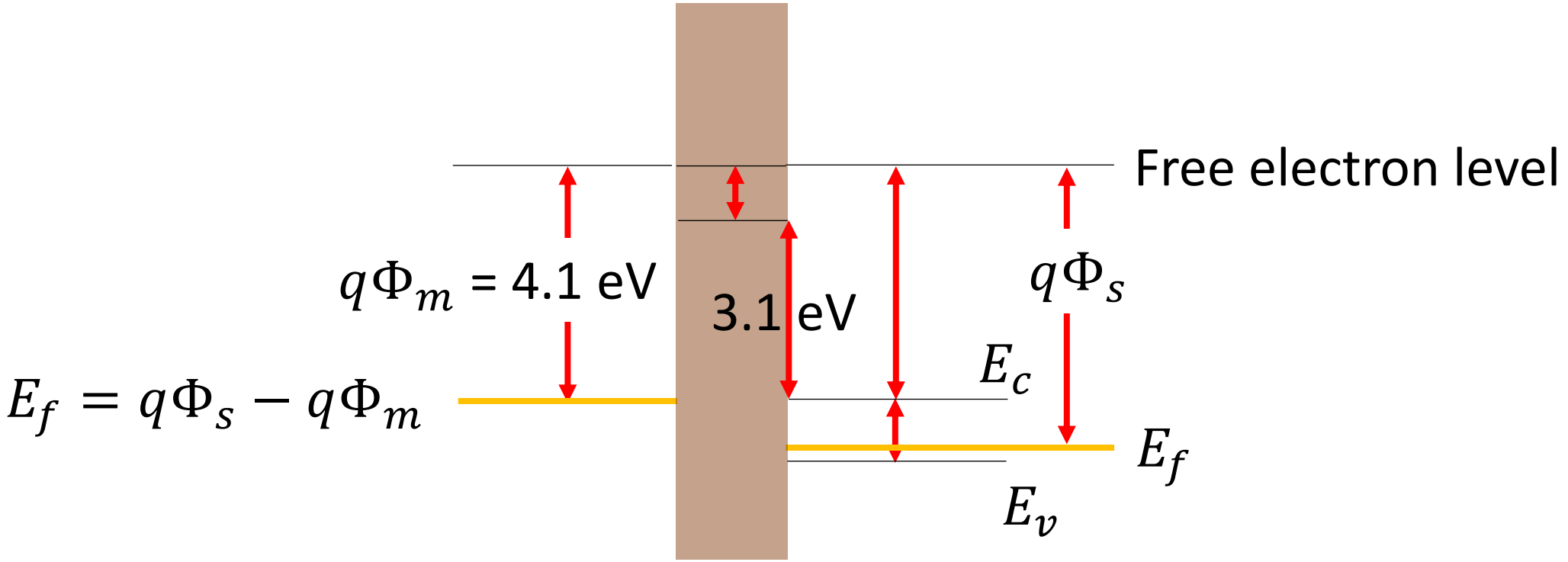
- The energy band at gate moves upward.
  - There is no energy difference.
  - It means that the energy band becomes flat.
  - This gate voltage is called the flatband voltage, V_{fb}.





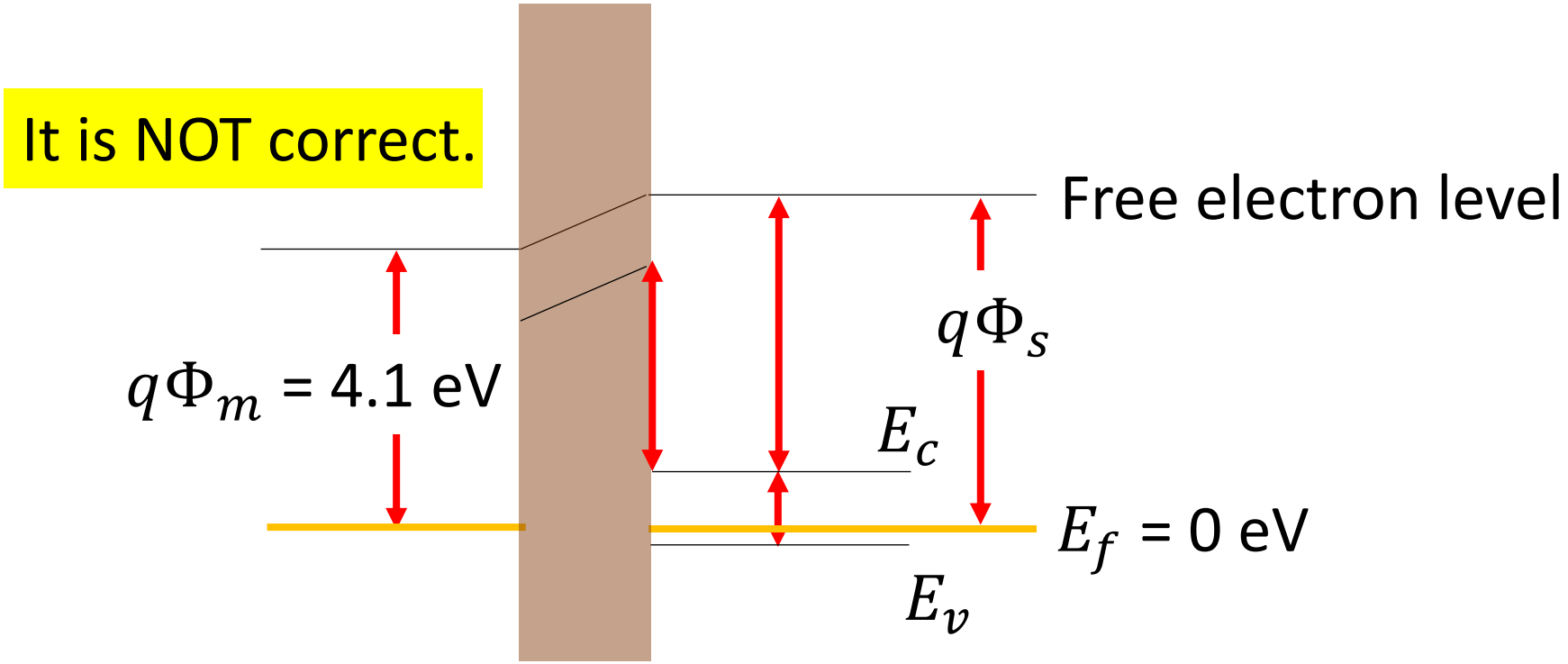
Draw the energy band diagram at $V_g = V_{fb}$.

- Since the energy band is flat, it is not difficult.
  - The electron energy barrier is 3.1 eV between the conduction bands of silicon and silicon dioxide.



Consider $V_g = V_{sub} = 0$ V, again.

- Non-zero electric field is found.
  - However, the energy difference, $q\Phi_s - q\Phi_m$, cannot be solely applied to the oxide layer. *Why?*



Thank you!